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PhD training needs a reboot in an AI world

As machines get better at data analysis and writing tasks, doctoral training must evolve to make the most of artificial-intelligence outputs. **By Alex Sen Gupta**

Since the release of the chatbot ChatGPT in late 2022, there has been frantic debate at universities about artificial intelligence (AI). These conversations have centred on undergraduate teaching – how to prevent cheating and how to use AI to improve learning. But a quieter, deeper disruption is unfolding in research, the other core activity of universities.

A doctoral education has long been seen as the pinnacle of academic training, an apprenticeship in original thinking, critical analysis and independent enquiry. However, that model is now under pressure. AI is not just another research tool; it is redefining what research is, how it is done and what counts as an original contribution.

Universities are mostly unprepared for the scale of disruption, with few having comprehensive governance strategies. Many academics remain focused on the failings of early generative AI tools, such as hallucinations (confidently stated but false information), inconsistencies and superficial responses. But AI models that were clumsy in 2023 are becoming increasingly fluent and accurate.

AI tools can already draft literature reviews, write sophisticated code with human guidance and even generate hypotheses when provided with data sets. 'Agentic' AI systems that can set their own sub-goals, coordinate tasks and learn from feedback represent another leap forwards. If the current trajectory continues, we're fast approaching a moment when much of the conventional PhD workflow can be completed, or at least be heavily supported, by machines.

Unanswered questions

This shift poses challenges for educators. What constitutes an original contribution becomes unclear when AI tools produce literature reviews, acquire and analyse data, and draft thesis chapters. Students might need to pivot from executing research tasks to framing questions and interrogating AI outputs.

To explore what the near future of research training might look like, I conducted a role play simulating a PhD student working with a hypothetical AI assistant. I used Claude, a leading AI system built by the firm Anthropic in San Francisco, California.

I fed the chatbot a detailed prompt (see Supplementary Information) describing a fictional AI research assistant called HALe – inspired by the AI character HAL 9000 from the science-fiction film *2001: A Space Odyssey*. I gave HALe capabilities that are already under development and are likely to improve in coming years. These include accessing external databases, integrating environmental and biological data, and performing advanced analyses autonomously. I then played the part of the student, asking questions and responding to the chatbot's replies. The dialogue was generated in a single, unedited session – offering a fictional, yet plausible, glimpse of how future doctoral research could unfold.

The simulated goal was to complete a PhD project investigating how extreme ocean temperatures affect marine species – an ambitious task involving data synthesis, statistical modelling and writing a paper for publication. In this fictional scenario, HALe didn't merely assist; it took initiative. It searched and extracted data from scientific literature, identified knowledge gaps, harmonized environmental and biological data sets, ran complex statistical analyses, interpreted the results, drafted a manuscript, suggested peer reviewers and even created an open-access data repository. The entire process, which would realistically take a student several months, played out in a short sequence of guided exchanges that might occupy just a few hours.

Although today's AI models cannot yet perform these tasks with anything approaching full autonomy, the simulation was grounded in what current systems can already do with human guidance. For example, ChatGPT, Claude and other state-of-the-art chatbots can draft credible literature reviews, propose hypotheses, suggest analytical approaches and generate code that – when reviewed and validated by a human – can process real data sets and produce meaningful outputs. They can even help to interpret statistical results and visualize findings. What struck me, while conducting this exercise, was how much of the conventional PhD process could now be driven and accelerated by AI. At times, it felt like working with a hyper-competent and astonishingly rapid research assistant. It was both exciting and unsettling.

Of course, this simulation reflects a particular kind of project – analytical, data-rich and computational in nature. Experimental or field-based PhD programmes, especially those that require collecting samples, laboratory work or interacting with other people or with the natural world, will remain less susceptible to full automation. But even in these areas of science, AI is likely to play a growing part in experimental design, autonomous data collection, literature synthesis and post-experiment analysis.

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New skills

This experience brought home how training in academic skills will need to be fundamentally reconsidered in an era of AI.

Certain proficiencies, such as data analysis and literature synthesis, might need to take a back seat so that researchers can focus on critical evaluation of AI-generated code and analyses, drawing conclusions from machine-processed data and maintaining oversight of increasingly automated workflows. Crucially, students will need to develop sufficient expertise to identify when AI systems produce plausible but incorrect outputs – which poses a dilemma because this requires the very skills that AI is starting to replace.

Postgraduate assessments will also need to evolve. A polished thesis or paper could mask a shallow understanding of the topic. Oral defences, live problem-solving and reflective commentary on research decisions might become more important than a written thesis. Supervisors will need to learn how to guide students to use AI tools critically and responsibly.

The structure of PhD programmes will need to be reconsidered. If it becomes possible to drastically compress months of PhD student labour, universities will need to decide whether to shorten doctoral programmes or expand their scope to tackle more ambitious, potentially interdisciplinary problems.

Most crucially, safeguarding student development is essential because AI reliance could lead to intellectual atrophy across a range of skills. Without regular practice, student abilities – such as critical reading, analytical reasoning and scientific writing – could weaken. Deliberate pedagogical intervention will be required to maintain critical thinking, because AI can become a cognitive crutch.

There are no easy answers, but we cannot simply wait and see. How we undertake research is radically changing. Clinging to conventional processes while ignoring transformative technologies risks our programmes becoming irrelevant and removes our agency for directing how AI will shape the future of research and knowledge creation.

Some institutions are beginning to respond. Many now issue AI-use guidelines, focusing mainly on academic integrity – for instance, requiring disclosure of AI assistance and restricting its use to support roles rather than substantive research writing. A few institutions, such as the University of Oxford, UK, Nanyang Technological University in Singapore and the Karolinska Institute in Stockholm, have released research-specific policies. Oxford offers AI and machine-learning competency courses for staff members and Nanyang mandates AI literacy for postgraduate students. Some universities are starting to provide campus-wide access to advanced AI systems, such as ChatGPT Edu at the University of New South Wales in Sydney, Australia. Other institutions don't have the resources to do this – risking far-reaching inequities in research capacity.

These early initiatives are important first steps, but they mostly treat AI as a powerful tool rather than a transformative force. They do not yet address what will happen when AI takes over much of the PhD workflow or when originality itself needs to be redefined.

Many academics remain sceptical about how much disruption AI might bring or how fast it will happen. Yet it's hard to imagine that AI's capabilities have reached their peak. I challenge any supervisor to test their current capabilities at first hand: pose a research question in your field to one of today's leading AI models. Ask it to identify new research directions, outline methodological approaches or even suggest analytical frameworks, and watch it execute complex analyses on real data. For those that don't use AI tools regularly, the sophistication of the response might be unsettling – a preview of the world that PhD students will inhabit.

AI is not going away. The question is whether we will shape its role in research training or let it shape us.

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